

01_analysis

Helix OTA — Requirement Analysis of additions/initial_research_01.md

Field	Value
Revision	1
Created	2026-06-08
Last modified	2026-06-08
Status	active
Status summary	Exhaustive requirement-by-requirement mapping of the operator-supplied draft additions/initial_research_01.md (the “Mender-wrap master plan”) against the current canonical corpus (docs/research/main_specs/) and the built server code (server/). Every line/section of the draft is decomposed into atomic requirements (REQ-01-NN), categorized, and assigned a coverage verdict (COVERED / PARTIAL / MISSING) with a citing file or an explicit gap. Conflicts with the locked decisions / ADRs are flagged with a recommended resolution.
Issues	The draft’s central thesis (wrap Mender) conflicts with locked decision D2 and is routed to ADR-0001 (still Proposed, not Accepted). Several draft submodules (go-common, helm-charts, vasic-digital/secrets, vasic-digital/auth) are UNVERIFIED / do not exist under those names (per additions_synthesis.md §6). HelixConstitution clause numbers cited throughout the corpus are UNVERIFIED.
Issues summary	Most of draft 01 is already absorbed into the corpus + code; the residual

Field	Value
	gaps are the dashboard, the OpenAPI multi-format export wiring, and a handful of telemetry/deployment specifics.
Fixed	N/A (initial revision of this per-draft analysis).
Fixed summary	First exhaustive single-draft analysis; supersedes the high-level treatment of draft 01 in <code>additions_synthesis.md</code> (Rev 1).
Continuation	Author <code>02_analysis.md</code> (draft 02) and <code>03_analysis.md</code> (draft 03 dir) to the same depth; fold the MISSING/PARTIAL items below into the named destination specs; re-verify Mender/Android claims live when ADR-0001 closes.

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1. purpose and method

Per operator mandate (every file in `additions/` is authoritative input that **MUST** be deeply analyzed; HelixConstitution §11.4.6 no-guessing, §11.4 anti-bluff — both UNVERIFIED clause numbers, carried from corpus convention), this document maps **`additions/initial_research_01.md`** in full. Method:

1. **Decompose** the draft into atomic requirements (REQ-01-NN), each tagged with its source section/line.
2. **Categorize** each (architecture / api / db / security / client / rollout / deployment / testing / docs / roadmap).
3. **Map** each to the current state: the canonical corpus under `docs/research/main_specs/` and the built Go server under `server/`. Cite the covering file or mark the gap.
4. **Flag conflicts** with the locked decisions / the five ADRs, with a recommended resolution (defer-to-ADR / reject / accept-with-modification).

Cross-reference inputs read for this analysis: the draft itself; `research/additions_synthesis.md` (Rev 1, prior high-level pass on drafts 01+02); `research/adr/adr-0001..0005`; the master design `00-master/2026-06-07-helix-ota-design.md`; `00-master/threat_model.md`; the 1.0.0-mvp specs

(api/, database/, security/, server/, client_android/, deployment/, tests/); and the built server (server/internal/api/server.go, server/internal/store/store.go, handlers).

2. coverage legend

- **COVERED** — the requirement is fully represented in a canonical spec and/or implemented in server/ code; citation given.
- **PARTIAL** — partially specified or specified-but-not-built (or built-but-not-spec'd); residual work named.
- **MISSING** — no canonical spec or code addresses it; gap named.
- **Conflict** column: – = none; otherwise the ADR / locked decision it collides with.

3. source decomposition summary

initial_research_01.md is a 592-line “master plan” whose thesis is: **wrap Mender** as the core engine, orchestrate it with a custom Go server, drive Android devices via UpdateEngine, and lay out a per-phase directory tree with multi-format export. It contributes: an OTA comparison matrix (§2.1), a three-tier decoupled architecture (§3), an MVP scope + version roadmap (§4), a /api/v1 REST sketch (§5.1), a first-cut PostgreSQL schema (§5.2), an Android client design (§5.3), an upload/verify flow (§5.4), containerisation (§5.5), security measures (§5.6), a submodule strategy (§6), a phase directory tree (§7), Mermaid diagrams (§3.1/§8), a testing strategy (§9), Constitution compliance (§10), kickoff steps (§11), and an export plan (§12).

The overwhelming majority is already absorbed into the corpus and/or built in code; the draft’s distinctive choices (Mender, gRPC, microservices-adjacent service split, Vault, Traefik+nginx, React dashboard, 90% coverage) are either already resolved as conflicts or are PARTIAL/MISSING.

4. requirement inventory

ReqID	Requirement (source §/line)	Category	Coverage
REQ-01-01	Universal, generic, decoupled OTA server+client embeddable into any OS; Android-first, later Linux/Windows/other (§1, L16–27)	architecture	COVERED — 00-master/2026-06-07-helix-ota-design.md (universality search OS-adapter); future-OS phases 1.X-linux/, 1.X-windows/other-os/
REQ-01-02	Guarantees: zero-corruption atomic verified revertible updates (§1, L23)	architecture/ security	COVERED — native A/B + dm-verity, 00-master D2; threat_model.md §4.3
REQ-01-03	Granular rollout control (5/10/30/.../100 in any steps) (§1, L24)	rollout	COVERED (as deferred) — 1.0.1-staged-rollout/README.md; design defers staged rollout to 1.0.1

ReqID	Requirement (source §/line)	Category	Coverage
REQ-01-04	Full telemetry for problem detection/ reporting (§1, L25)	client/ rollout	COVERED — server/ internal/api/ handlers_client.go (telen 1.0.0-mvp/server/ telemetry_processing.mo
REQ-01-05	Enterprise scalability single board → millions of devices (§1, L26)	architecture/ deployment	PARTIAL — horizontal scali noted in deployment/ overview.md; HPA threshold UNVERIFIED, derived from tests later
REQ-01-06	Deep pluggability: new OS backends with minimal effort (§1, L27)	architecture	COVERED — OS-adapter se 00-master design §4
REQ-01-07	Deep research of existing OTA solutions; build-vs-wrap decision (§2, L31–35)	docs/ architecture	COVERED — research/ ota_landscape_report.mo research/stacks/*; decisio routed to ADR-0001
REQ-01-08	OTA comparison matrix (Mender/RAUC/ SWUpdate/OSTree/AOSP/Go-OTA) (§2.1, L38–46)	docs	COVERED — expanded with evidence in research/ ota_landscape_report.mo per-stack notes
REQ-01-09	Conclusion: wrap/encapsulate Mender as the core engine (§2, L47–52)	architecture	PARTIAL — ADR-0001 lists Mender as Option B but statu Proposed , not Accepted; loca D2 biases to minimal-wrappi AOSP-native-only
REQ-01-10	Three-tier split: OTA Server (Go) / Client SDK (per-OS) / Dashboard (Web UI) (§3, L57–62)	architecture	COVERED — 00-master d §4 (server/client/dashboard ti
REQ-01-11	All components containerised, deployed via K8s/Compose using vasic-digital/ containers (§3, L63)	deployment	COVERED — 1.0.0-mvp/ deployment/overview.md, docker-compose.mvp.yml, kubernetes/
REQ-01-12	High-level architecture diagram (Mermaid) (§3.1, L67–99)	docs	COVERED — 1.0.0-mvp/ architecture/diagrams/ ; design embeds
REQ-01-13	Mermaid diagram references Mender Server (wrapped) as a core sub-service (§3.1 L74; §3.2 L103–104)	architecture	PARTIAL — same Mender dependency as REQ-01-09; diagrams in corpus are native control-plane, not Mender- wrapping
REQ-01-14	PostgreSQL for relational state (Mender + custom tables) (§3.2 L105)	db	COVERED — 1.0.0-mvp/ database/schema.md; pgx- backed Repository is the production target (store/ store.go)
REQ-01-15	MinIO S3-compatible artifact storage, encrypted at rest (§3.2 L106)	deployment/ security	COVERED — deployment/ minio_setup.md; Storage design D7

ReqID	Requirement (source §/line)	Category	Coverage
REQ-01-16	Telemetry & Analytics service ingests device reports → dashboard (§3.2 L107)	rollout	PARTIAL — ingestion built (handlers_client.go) + telemetry_processing.mo analytics/aggregation + dashboard surfacing not built
REQ-01-17	Android client = privileged system app/service; polls, downloads, verifies (hash+sig), writes inactive slot, reboots via UpdateEngine (§3.2 L108; §5.3)	client	COVERED — client_and integration_guide.md, build_integration.md, update_engine_integrati
REQ-01-18	MVP goal: upload, distribute, install OTA to one or all devices (no partial rollouts) (§4.1 L116–118)	rollout	COVERED — deployments all strategy; handlers_deployment.go, store.Deployment.Strate
REQ-01-19	Secure upload of OTA zip + hash file via dashboard, admin auth (§4.1 L123)	api/security	COVERED (server side) — handlers_artifact.go POST artifacts/upload, role-ga (Operator/Admin); dashboard MISSING (see REQ-01-49)
REQ-01-20	Server-side artifact validation: checksum, signature, metadata extraction (§4.1 L124; §5.1.1; §5.4)	api/security	COVERED — ota-artifact validator wired in server artifact_validation.md; signature verify hardened (co da4312b)
REQ-01-21	Android client polls every X minutes, downloads latest approved artifact (§4.1 L125; §5.3.2)	client	COVERED — integration_guide.md (WorkManager 15 min + jitter configurable) — note poll ca resolved (C3)
REQ-01-22	Atomic A/B install via UpdateEngine (seamless, no data loss) (§4.1 L126)	client	COVERED — update_engine_integrati (applyPayload, Virtual A/B
REQ-01-23	Verify after download (hash) and after install (post-boot check) (§4.1 L127)	client/ security	COVERED — integration_guide.md ver before-apply + post-boot hea telemetry
REQ-01-24	Device registration + basic inventory (board ID, OS version, build fingerprint) (§4.1 L128; §5.1.2; §5.3.1)	api/client	COVERED — handlers_device.go POST devices/register; store.Device (HardwareID Model, OSType, OSVersion, fingerprint via Metadata)
REQ-01-25	Simple “deploy to all” capability (§4.1 L129)	rollout	COVERED — store.Deployment all-target strategy; handlers_deployment.go
REQ-01-26	Rollback NOT required for MVP (A/B fallback remains) (§4.1 L130)	rollback	COVERED — matches lock (automatic A/B boot-failure rollback only in MVP)
REQ-01-27			

ReqID	Requirement (source §/line)	Category	Coverage
	Telemetry: success/failure reported, visible in dashboard (§4.1 L131)	client/ rollout	PARTIAL — reporting COVERED; dashboard visible MISSING (no dashboard)
REQ-01-28	MVP exclusions list (no staged/% rollout, no rich dashboards, no user rollback, no multi-OS, no delta) (§4.1 L134)	roadmap	COVERED — matches design scope + ADR-0005 (delta deferral) + C6
REQ-01-29	Phase 1.0.1 — staged rollouts (5/10/30/.../100), canary groups, real-time monitoring, auto pause/rollback on threshold (§4.2 L136–143)	rollout/ roadmap	COVERED (as future) — 1.0.1 staged-rollout/README.m rollout-engine seed from draft routed there
REQ-01-30	Phase 1.0.2 — user-controlled rollback + delta updates (snapshot, dashboard/recovery rollback, Mender/custom delta) (§4.3 L145–151)	roadmap/ rollback	PARTIAL — rollback deferral COVERED (C6); delta routing ADR-0005 (phase placement open); a dedicated 1.0.2- rollback/ phase dir is MISSING
REQ-01-31	Future OS expansions: 2.0.0 Linux (RAUC/swupdate), 3.0.0 Windows, 4.0.0 universal orchestrator (§4.4 L153–161)	roadmap	COVERED — 1.X-linux/, windows/, 1.X-other-os/ README outlines
REQ-01-32	Server design: new OS = new client lib, server unchanged (new endpoints only for OS-specific metadata) (§4.4 L161)	architecture	COVERED — OS-adapter section 00-master design §4
REQ-01-33	All endpoints versioned /api/v1/; JWT Bearer auth via Helix IAM (§5.1 L169)	api/security	COVERED — server.go APIBasePath group, authMiddleware, token signing endpoints.md §7
REQ-01-34	POST /artifacts/upload (SHA256 check, RSA/ECDSA sig verify, metadata extract) (§5.1.1 L173–176)	api/security	COVERED — handlers_artifact.go; signature alg is ed25519 in code (server.go ed25519.PublicKey), not RSA ECDSA per draft
REQ-01-35	GET /artifacts/{id} metadata retrieval (§5.1.1 L177)	api	COVERED — handlers_artifact.go GET artifacts/:artifactId
REQ-01-36	POST /devices/register (serial, fingerprint, IP) → device token (§5.1.2 L181)	api	COVERED — handlers_device.go; return device-scoped token
REQ-01-37	GET /devices/{id}/status (firmware version, last check-in) (§5.1.2 L182)	api	COVERED — handlers_device.go GET devices/:deviceId/status
REQ-01-38	POST /deployments (artifact + target group all) → deployment ID (§5.1.3 L186)	api	COVERED — handlers_deployment.go bound to release , not raw artifact — see REQ-01-43)
REQ-01-39	GET /deployments/{id} status (§5.1.3 L188)	api	COVERED — handlers_deployment.go deployments/:deploymentId
REQ-01-40		api/client	

ReqID	Requirement (source §/line)	Category	Coverage
	GET /client/update? device_token¤t_build → 204 or 200 (URL, token, expiry, SHA256) (§5.1.4 L192–194)		COVERED — handlers_client.go GET . client/update; 204/200 co in endpoints.md §12.1
REQ-01-41	POST /telemetry/report JSON (status, error code, logs) → PostgreSQL (§5.1.5 L198)	api	COVERED — handlers_client.go POST client/telemetry; AppendTelemetry (path diff client/telemetry)
REQ-01-42	OpenAPI schema at phase/1.0.0-mvp/ api/openapi.yaml (§5.1 L200)	api/docs	COVERED — 1.0.0-mvp/a openapi.yaml exists
REQ-01-43	DB: helix_ota.releases (artifact_id, os_type, board_name, version, rollout_percentage, rollout_group) (§5.2 L208–218)	db	PARTIAL — releases table COVERED in schema.md an store.Release, but rollout_percentage/ rollout_group columns are deferred to 1.0.1 (no staged in MVP)
REQ-01-44	DB: helix_ota.telemetry_events (device_id, deployment_id, event_type, payload JSONB) (§5.2 L221–228)	db	COVERED — schema.md § telemetry_events; store.TelemetryRecord
REQ-01-45	DB: helix_ota.rollback_history (from/to release, triggered_by) (§5.2 L231–238)	db	COVERED (as deferred) — schema.md continuation note rollback_history added in (matches C6)
REQ-01-46	Full DDL + migration scripts in database/migrations/ (§5.2 L241)	db	COVERED — 1.0.0-mvp/ database/migrations/ exi
REQ-01-47	Android client repo helix-ota-client- android (Kotlin/Java + Go wrapper option) (§5.3 L243–245)	client	PARTIAL — corpus names t NEW submodule ota-andro agent (KMP), not helix-ot client-android; repo not y created (UNVERIFIED until exists)
REQ-01-48	Client: registration (ro.serialno,ro.build.fingerprint), WorkManager polling, download+verify, applyPayload, callback→telemetry, exp- backoff retry, A/B fallback (§5.3.1–3 L246–254)	client	COVERED — integration_guide.md (al these)
REQ-01-49	Client permissions: privileged system app / platform-signed; INSTALL_PACKAGES not allowed; UpdateEngine needs RECOVERY/system UID; built into AOSP (§5.3 L256)	client	COVERED — build_integration.md (sy UID vs priv-app, platform sig SELinux)
REQ-01-50	Update communication sequence diagram (§5.3 L260–282)	docs	COVERED — architectur diagrams/ sequence + update_engine_integrati
REQ-01-51	Upload→verify flow: admin login → select signed zip + .sha256 → multipart	api/security	COVERED — artifact_validation.md

ReqID	Requirement (source §/line)	Category	Coverage
	upload → server verifies → extract META-INF/com/android/metadata → store MinIO → record release @100% (§5.4 L284–296)		handlers_artifact.go + multipart.go
REQ-01-52	Validation pipeline doc with decision tables at server/artifact_validation.md (§5.4 L297)	docs	COVERED — 1.0.0-mvp/server/artifact_validation.md
REQ-01-53	docker-compose.ota.yml services: ota-server, mender-server , postgres, minio, traefik (Let's Encrypt) , dashboard (React via nginx) (§5.5 L304–310)	deployment	PARTIAL — docker-compose.mvp.yml has ota-server/postgres/minio; mender-server dropped (ADR-0001); traefik nginx reverse-proxy choice (Caddy vs nginx vs http3-from-Go, ADR-0004); dashboard server not wired
REQ-01-54	Kubernetes manifests under kubernetes/ for production scaling (§5.5 L311)	deployment	COVERED — 1.0.0-mvp/deployment/kubernetes/
REQ-01-55	Secrets (DB pw, JWT secret, signing keys) via HashiCorp Vault or K8s Secrets / vasic-digital/secrets submodule (§5.5 L313)	deployment/security	PARTIAL — deployment/overview.md explicitly rejects invented Vault submodule ; container/native secret mechanisms; vasic-digital/secrets UNVERIFIED/nonexistent
REQ-01-56	Security: artifact signing (private key sign, public key in client keystore) (§5.6 L317)	security	COVERED — signing_verification.md; key_management.md; ed25519 pubkey in server config
REQ-01-57	Security: TLS 1.3 mandatory (§5.6 L318)	security	COVERED — transport_security.md; threat_model §4
REQ-01-58	Security: device auth via mutual TLS or JWT bound to hardware ID (Android KeyStore) (§5.6 L319)	security	PARTIAL — JWT + KeyStore binding COVERED; mTLS “evaluated”, not committed MVP (threat_model §4.x reserved “does not commit to mTLS”)
REQ-01-59	Security: rollback/downgrade protection via AVB + A/B boot control (bootloader version check) (§5.6 L320)	security	COVERED — threat_model §4.3 (AVB rollback-index / a downgrade)
REQ-01-60	Security: audit logs for every admin action (§5.6 L321)	security	PARTIAL — audit_logs table planned in schema.md continuation; not built in server (no audit middleware in hand)
REQ-01-61	New reusable components as public repos on GitHub+GitLab (HelixDevelopment), added as submodules (§6 L327–328)	docs/architecture	COVERED — pre-authorized public repo creation (locked) master/submodule_reuse_map.md

ReqID	Requirement (source §/line)	Category	Coverage
REQ-01-62	New submodules: helix-ota-server, -client-android, -dashboard, -common, -client-linux, -client-windows (§6.1 L332–339)	architecture	PARTIAL — corpus uses NE ota-* repos (ota-protocol, artifact-validator, ota-update-engine-bridge, ot-telemetry-schema, ota-android-agent); naming di and repos not yet created (UNVERIFIED)
REQ-01-63	Existing submodule deps: containers, go-common, auth, helm-charts, HelixConstitution (§6.2 L343–352)	architecture	PARTIAL — containers, HelixConstitution verifie common, helm-charts, vasi digital/auth UNVERIFIED do not exist under those nam (additions_synthesis.md)
REQ-01-64	Contribute generic staged-rollout/canary library back to containers via PRs (§6.2 L354)	rollout/ architecture	PARTIAL — rollout engine i OS-agnostic submodule sean (1.0.1 + design); “contribute containers” placement UNVERIFIED
REQ-01-65	Per-phase directory tree (architecture/api/database/security/server/client_android/tests/deployment/exports) (§7 L362–421)	docs	COVERED — 1.0.0-mvp/ mirrors this (re-rooted under research/main_specs/)
REQ-01-66	Convert all Markdown → PDF/HTML/DOCX via pandoc into exports/ (§7 L424; §12 L572–580)	docs	COVERED — 00-master/export_pipeline.md + scripts/export_docs.sh; HTML/DOCX/SVG/PNG ve PDF needs LaTeX engine (SKIPPED until tooling)
REQ-01-67	Database ER diagram (Mermaid) (§8.1 L434–479)	docs/db	COVERED — database/schema.md ER + architect diagrams/
REQ-01-68	Staged rollout engine flow diagram (§8.2 L483–492)	docs/rollout	COVERED (future) — folde 1.0.1-staged-rollout/
REQ-01-69	Container architecture deployment diagram (traefik→ota-server→mender/db/minio) (§8.3 L496–506)	docs/ deployment	PARTIAL — deployment dia exists but without mender/tra specifics (see REQ-01-53)
REQ-01-70	Diagrams as .mmd + SVG/PNG via mermaid-cli (§8 L430, L508)	docs	COVERED — export_pipeline.md (mmd verified)
REQ-01-71	Unit tests (Go): validation, rollout, telemetry; Android: downloader, hash, UpdateEngine (mocked). Coverage ≥ 90% (§9.1 L514–519)	testing	PARTIAL — four-layer strate COVERED (tests/test_strategy.md); covera target resolved by Constitutio four-layer + mutation, not a 90% (C8)
REQ-01-72	Integration tests: API via httptest + real PG/MinIO (testcontainers-go); client-server emulator sim (§9.2 L521–524)	testing	COVERED — test_strategy.md; built te httptest (handlers_*_test.
REQ-01-73		testing	

ReqID	Requirement (source §/line)	Category	Coverage
	E2E on Orange Pi 5 Max: flash, upload, deploy, verify download/install/reboot/report; A/B fallback by corrupting slot (§9.3 L526–532)		COVERED — test_strategy.md §e2e (Orange Pi 5 Max)
REQ-01-74	Performance/load: 100k devices polling+downloading via k6 ; K8s HPA horizontal scale (§9.4 L534–537)	testing/deployment	COVERED — test_strategy.md §10 (k6 thresholds) UNVERIFIED (REQ-01-05)
REQ-01-75	Constitution compliance: 2-reviewer PRs + CODEOWNERS, full docs, no-untested-merge CI, modularity, OWASP+gosec+Dependabot, Conventional Commits (§10 L543–554)	docs/testing	PARTIAL — principles reflected across corpus + test_strategy.md; concretized (GitHub Actions, CODEOWNERS, gosec, Dependabot) not present in
REQ-01-76	Kickoff steps: clone, add submodules, compose up, build upload→check endpoints first, AOSP client, tests alongside (§11 L558–568)	roadmap	COVERED — sequencing consistent with corpus; upload+client endpoints already built
REQ-01-77	Multi-format export distribution (PDF/HTML/DOCX, pre-rendered SVG/PNG) (§12 L572–580)	docs	COVERED — see REQ-01-06 (PDF gated on LaTeX)
REQ-01-78	OTA Dashboard = React SPA (admin login, upload, deploy button, telemetry view) (§3 L60; §5.4; §5.5 L310)	client/architecture	MISSING — no dashboard source code in corpus/server/; desired names React + *-React bricks; no helix-ota-dashboard source repo, or implementation exists
REQ-01-79	gRPC API surface (implied by §3.1 “REST / gRPC API” box, L70)	api	PARTIAL — REST is the mandated primary surface; gRPC optional/internal only (C4); not built
REQ-01-80	Reverse proxy with Let’s Encrypt TLS termination (Traefik) (§5.5 L309)	deployment	PARTIAL — proxy choice Open (Caddy vs nginx vs http3-forwarding) pending ADR-0004 QUICK spike

5. conflicts with locked decisions and adrs

#	Draft requirement	Conflicts with	Recommended resolution
K1	REQ-01-09/13 Wrap Mender as the core engine	Locked D2 (native A/B + custom Go control plane, minimal wrapping); ADR-0001 still <i>Proposed</i>	Defer to ADR-0001. Do not pre-commit Mender. Mender’s weak/unofficial Android support (synthesis §6) materially lowers its score; AOSP-

#	Draft requirement	Conflicts with	Recommended resolution
			native-only is the third option and the bias.
K2	REQ-01-79 / §3.1 gRPC as a co-primary API	Mandated stack: Gin + REST primary , HTTP/3→HTTP/2, Brotli (C4)	Accept-with-modification. REST is the compatibility surface; gRPC optional/internal only, post-MVP.
K3	REQ-01-53 microservice-style split (separate mender-server, telemetry service, auth service containers)	ADR-0003 modular monolith for MVP (C2)	Accept-with-modification. One Go deployable with internal package seams; service extraction at scale.
K4	REQ-01-55 HashiCorp Vault / vasic-digital/secrets submodule for secrets	deployment/overview.md explicitly rejects an invented Vault submodule; no such submodule	Reject the Vault submodule; use container-native secrets + security/config bricks. Vault only if a catalogue brick can't meet need (§11.4.74).
K5	REQ-01-63 existing submodules go-common, helm-charts, vasic-digital/auth	Verified catalogue (synthesis §6); these names do not exist	Reject guessed names. Use auth, security, database, Storage, config, observability, etc.
K6	REQ-01-71/§9.1 flat 90% coverage target	C8 — Constitution §1 four-layer + mutation immunity supersedes a single percentage	Accept-with-modification. Per-component targets; floor ≥90% on safety-critical paths (signing, apply, rollout-gate).
K7	REQ-01-34 signature alg RSA/ECDSA	Code + ADR-0002 trust model use ed25519 (server.go)	Accept-with-modification. ed25519 is the canonical artifact-signing alg; revisit cert/alg agility under ADR-0002.
K8			

#	Draft requirement	Conflicts with	Recommended resolution
K9	REQ-01-21 poll cadence “every X minutes” (unspecified)	Locked default 15 min + jitter, configurable (C3)	Resolved — 15 min + jitter is canonical; draft’s constant is non-binding.
	REQ-01-30 user-controlled rollback in 1.0.2 (early)	Locked C6 — end-user/multi-version rollback deferred past MVP	Resolved/deferred. MVP = automatic A/B fallback only; user rollback = 1.0.1+.
K10	REQ-01-31 OS-expansion numbering 2.0.0/3.0.0/4.0.0	Corpus uses 1.X-linux/1.X-windows/1.X-other-os outlines	Accept-with-modification (cosmetic): adopt corpus 1.X numbering or re-number when phases are promoted.

6. unverified claims in the draft

- **UNVERIFIED** — Mender comparison-matrix cells: “client C++/Go”, Android support marked “?*” (§2.1). Mender’s Android support is weak/unofficial; must be confirmed live in ADR-0001 (synthesis §6).
- **UNVERIFIED** — That vasic-digital/secrets, go-common, helm-charts, vasic-digital/auth exist (§5.5, §6.2). Per synthesis §6 they do not exist under those names.
- **UNVERIFIED** — “Mender server already provides APIs for deployments/devices/inventory we will leverage as the core” (§2 L51) — Mender adoption itself is unsettled (ADR-0001).
- **UNVERIFIED** — Claim that the OTA package always carries META-INF/com/android/metadata extractable for compatibility (§5.4 L292): depends on the RK3588 BSP shipping AOSP A/B vs a Rockchip proprietary OTA — flagged as the top hardware gate in build_integration.md.
- **UNVERIFIED** — HelixConstitution clause numbers (§11.4.x) cited across the corpus (carried convention; README and standards both flag UNVERIFIED).
- **Marketing, not evidence** — the draft’s framing (“ultimate”, “nano-level”, “leaves no detail unaddressed”, §1/§13) is stripped from canonical docs per §7.1 no-bluff.

7. gap rollup (missing and partial)

MISSING (no spec, no code):

- **REQ-01-78 — React OTA Dashboard.** No dashboard spec, repo, or implementation. This is the single largest true gap in draft 01 vs current state. MVP-relevant (the draft’s entire upload/deploy/telemetry-view UX assumes it).

PARTIAL (highest-priority, MVP-relevant first):

- **REQ-01-09 / REQ-01-13 — Mender wrap decision** (ADR-0001 still *Proposed*). Blocks the deployment topology, the container set, and diagrams. Highest architectural gap.
- **REQ-01-60 — Audit logging.** `audit_logs` table is planned but unbuilt; no audit middleware in handlers. Security/compliance-relevant for MVP.
- **REQ-01-53 / REQ-01-69 / REQ-01-80 — Deployment topology & reverse proxy.** mender-server dropped; proxy choice (Caddy/nginx/http3-Go) open pending ADR-0004; dashboard service unwired; container diagram stale.
- **REQ-01-16 / REQ-01-27 — Telemetry analytics + dashboard visibility.** Ingestion built; aggregation/analytics and any UI surfacing not built.
- **REQ-01-43 — releases.rollout_percentage / rollout_group columns.** Deferred to 1.0.1 (correct), but the draft asks for them at MVP; track for the staged-rollout migration.
- **REQ-01-55 — Secrets management** (Vault rejected; container-native path chosen but not wired/verified).
- **REQ-01-62 / REQ-01-47 — NEW ota-* submodule repos** specified but **not yet created** (UNVERIFIED until they exist); naming differs from the draft's `helix-ota-*`.
- **REQ-01-75 — CI/governance config** (GitHub Actions, CODEOWNERS, gosec, Dependabot) described but absent from the repo.
- **REQ-01-30 — 1.0.2-rollback/ phase dir** not created (delta phase placement open, ADR-0005).
- **REQ-01-05 — HPA/resource sizing** UNVERIFIED (derive from load tests).
- **REQ-01-66 / REQ-01-77 — PDF export** gated on a LaTeX engine (HTML/DOCX/SVG/PNG verified).
- **REQ-01-58 — mTLS device auth** evaluated-only; deferred to 1.0.1+ hardening.
- **REQ-01-79 — gRPC** optional/internal only; not built (by design).
- **REQ-01-64 — rollout-engine repo home** (“contribute to containers”) undecided.

Resolved-by-prior-decision (not gaps, recorded for completeness): REQ-01-21 (C3 poll cadence), REQ-01-26/30 (C6 rollback scope), REQ-01-71 (C8 coverage), REQ-01-34/56 (ed25519 vs RSA/ECDSA via ADR-0002), REQ-01-31 (1.X numbering).